Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has an opposite side of 5 and hypotenuse 13. Find $\sin\theta$ and the length of the adjacent side.

Adjacent $= \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. $\sin\theta = \frac{5}{13}$.

2 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has an opposite side of 15 and hypotenuse 17. Find $\cos\theta$.

Adjacent $= \sqrt{17^2 - 15^2} = \sqrt{289 - 225} = \sqrt{64} = 8$. $\cos\theta = \frac{8}{17}$.

3 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has an adjacent side of 7 and hypotenuse 25. Find $\sin\theta$ and $\tan\theta$.

Opposite $= \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. $\sin\theta = \frac{24}{25}$, $\tan\theta = \frac{24}{7}$.

4 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, the two legs are 20 and 21, and the hypotenuse is 29. Find $\sin\theta$ and $\cos\theta$ for the angle θ opposite the side of length 20.

$\sin\theta = \frac{20}{29}$, $\cos\theta = \frac{21}{29}$.

5 **[Foundational]** • **Target: Special angles**

State the exact value of $\sin(30^\circ)$.

$\sin(30^\circ) = \frac{1}{2}$.

6 **[Foundational]** • **Target: Special angles**

State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.

$\sin(90^\circ) = 1$ and $\cos(0^\circ) = 1$.

7 **[Foundational]** • **Target: Converting between radians and degrees**

Convert 180° to radians.

$180^\circ = \pi$ radians.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Converting between radians and degrees

Convert $\frac{5\pi}{6}$ radians to degrees.

$\frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ$.

9 [Intermediate] • Target: The Pythagorean identity

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is acute, $\cos\theta = \frac{4}{5}$.

10 [Intermediate] • Target: The Pythagorean identity

Given $\cos\theta = \frac{5}{13}$ with θ acute, find $\sin\theta$.

$\sin^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\sin\theta = \frac{12}{13}$.

11 [Intermediate] • Target: Graphs: amplitude and period

State the amplitude and period of $y = 5\sin(x)$.

Amplitude: 5. Period: 2π.

12 [Intermediate] • Target: Graphs: amplitude and period

A sine graph has amplitude 4 and period π. Write its equation in the form $y = A\sin(Bx)$.

$A = 4$. $B = \frac{2\pi}{\pi} = 2$. So $y = 4\sin(2x)$.

13 [Intermediate] • Target: Graphs: phase shift and vertical shift

The graph of $y = \sin(x)$ is shifted $\frac{\pi}{3}$ to the right and 2 units up. Write the new equation.

$y = \sin\left(x - \frac{\pi}{3}\right) + 2$.

14 [Intermediate] • Target: Graphs: phase shift and vertical shift

State the vertical shift and midline of $y = 3\cos(x) + 5$.

Vertical shift: up 5. Midline: $y = 5$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Graphs: phase shift and vertical shift • Cross-Lesson: 7.01 Integration & Applications

The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the left. Write the new equation.

$y = \cos\left(x + \frac{\pi}{4}\right)$.

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16 [Challenging] • Target: Reference angles in other quadrants • Cross-Lesson: 8.01 Exponentials & Logarithms

Find $\sin(150^\circ)$ using the reference angle.

The reference angle is 30°; in Quadrant II, sine is positive. $\sin(150^\circ) = \sin(30^\circ) = \frac{1}{2}$.

17 [Challenging] • Target: Complementary angle relationships • Cross-Lesson: 9.01 Statistics & Probability

Given $\sin(20^\circ) \approx 0.342$, use the relationship $\sin\theta = \cos(90^\circ - \theta)$ to find $\cos(70^\circ)$.

$\cos(70^\circ) = \sin(20^\circ) \approx 0.342$.

18 [Challenging] • Target: The unit circle: coordinates of key points • Cross-Lesson: 1.01 Algebra & Equations

State the coordinates $(\cos\theta, \sin\theta)$ of the point on the unit circle at $\theta = 60^\circ$.

$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

19 [Challenging] • Target: Word problems: angles of elevation • Cross-Lesson: 2.01 Functions & Transformations

This question has two parts. A drone hovers directly above a landmark. From a point 200 meters from the base of the landmark, the angle of elevation to the drone is 35°.

a Write an equation relating the drone's height h to the given information, using the tangent ratio. $\tan(35^\circ) = \frac{h}{200}$, so $h = 200\tan(35^\circ)$.

b Find h, to the nearest meter, given $\tan(35^\circ) \approx 0.700$. $h \approx 200(0.700) = 140$ meters.

(a) $\tan(35^\circ) = \frac{h}{200}$, so $h = 200\tan(35^\circ)$.; (b) $h \approx 200(0.700) = 140$ meters.

20 [Challenging] • Target: Word problems: angles of elevation • Cross-Lesson: 3.01 Sequences & Series

This question has two parts. A ladder 10 meters long leans against a wall, making an angle of 65° with the ground.

a Write an equation for the height h the ladder reaches up the wall, using the sine ratio. $\sin(65^\circ) = \frac{h}{10}$, so $h = 10\sin(65^\circ)$.

b Find h, to the nearest tenth of a meter, given $\sin(65^\circ) \approx 0.906$. $h \approx 10(0.906) = 9.1$ meters.

(a) $\sin(65^\circ) = \frac{h}{10}$, so $h = 10\sin(65^\circ)$.; (b) $h \approx 10(0.906) = 9.1$ meters.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In the right triangle shown, angle θ has opposite side 8 and hypotenuse 17. Find $\sin\theta$.

a. $\frac{17}{8}$

b. $\frac{8}{15}$

c. $\frac{15}{17}$

d. $\frac{8}{17}$ ✓

$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{8}{17}$.

22 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In a right triangle, angle θ has opposite side 9 and hypotenuse 41. Find $\cos\theta$.

a. $\frac{9}{41}$

b. $\frac{9}{40}$

c. $\frac{41}{40}$

d. $\frac{40}{41}$ ✓

Adjacent $= \sqrt{41^2 - 9^2} = 40$; $\cos\theta = \frac{40}{41}$.

23 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In a right triangle, angle θ has adjacent side 20 and hypotenuse 29. Find $\sin\theta$.

a. $\frac{21}{20}$

b. $\frac{20}{29}$

c. $\frac{21}{29}$ ✓

d. $\frac{29}{21}$

Opposite $= \sqrt{29^2 - 20^2} = 21$; $\sin\theta = \frac{21}{29}$.

24 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In the right triangle shown, the two legs are 9 and 40, and the hypotenuse is 41. Find $\sin\theta$ and $\cos\theta$ (where θ is opposite the side of length 9).

a. $\sin\theta = \frac{9}{40}$, $\cos\theta = \frac{40}{9}$

b. $\sin\theta = \frac{40}{41}$, $\cos\theta = \frac{9}{41}$

c. $\sin\theta = \frac{41}{9}$, $\cos\theta = \frac{41}{40}$

d. $\sin\theta = \frac{9}{41}$, $\cos\theta = \frac{40}{41}$ ✓

Opposite $= 9$, adjacent $= 40$, hypotenuse $= 41$.

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25 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In a right triangle, angle θ has opposite side 7 and adjacent side 24. Find $\tan\theta$.

a. $\frac{24}{7}$

b. $\frac{24}{25}$

c. $\frac{7}{25}$

d. $\frac{7}{24}$ ✓

$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{7}{24}$.

26 [Foundational] • Target: MCQ — Graphs: amplitude and period

A cosine graph has amplitude 3 and period 4π. Write its equation in the form $y = A\cos(Bx)$.

a. $y = 3\cos(4\pi x)$

b. $y = 4\pi\cos(3x)$

c. $y = 3\cos\left(\frac{x}{2}\right)$ ✓

d. $y = 3\cos(2x)$

$B = \frac{2\pi}{\text{period}} = \frac{2\pi}{4\pi} = \frac{1}{2}$.

27 [Foundational] • Target: MCQ — Graphs: amplitude and period

The graph shows $y = A\sin(x)$ for some A. Find A by reading the graph's amplitude.

a. $A = 1$

b. $A = 2\pi$

c. $A = 4$

d. $A = 2$ ✓

The curve's peak height (amplitude) is 2.

28 [Foundational] • Target: MCQ — Graphs: phase shift and vertical shift

The graph shows $y = \sin(x)$ shifted vertically. Find the new midline by reading the graph.

a. $y = 0$

b. $y = 2$ ✓

c. $y = 1$

d. $y = -2$

The curve oscillates evenly about $y = 2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: MCQ — The unit circle: coordinates of key points

The unit circle shown marks a point at angle $\theta = 60^\circ$. What are its coordinates?

a. $\left(\frac{\sqrt{3}}{2}, 0.5\right)$

b. $(0.5, 0.5)$

c. $(1, 0)$

d. $\left(0.5, \frac{\sqrt{3}}{2}\right)$ ✓

$\cos(60^\circ) = 0.5, \sin(60^\circ) = \frac{\sqrt{3}}{2}$.

30 [Intermediate] • Target: MCQ — Word problems: angles of elevation

A drone hovers directly above a landmark. From a point 150 meters from the landmark's base, the angle of elevation to the drone is 40°. Find the drone's height, to the nearest meter.

a. 150

b. 196

c. 126 ✓

d. 115

$h = 150\tan(40^\circ) \approx 126$ m.

31 [Intermediate] • Target: MCQ — Word problems: angles of elevation

A ladder leans against a wall, making an angle of 70° with the ground. If the ladder shown is 12 meters long, find how high up the wall it reaches, to the nearest meter.

a. 4

b. 12

c. 8

d. 11 ✓

$h = 12\sin(70^\circ) \approx 11$ m.

32 [Intermediate] • Target: MCQ — Word problems: angles of elevation

From the top of an 80 m cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest meter.

a. 37

b. 80

c. 172 ✓

d. 189

$\tan(25^\circ) = \frac{80}{d} \Rightarrow d = \frac{80}{\tan(25^\circ)} \approx 172$ m.

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33 **[Intermediate]** • **Target: MCQ — Right-triangle trigonometric ratios**

In a right triangle, angle θ has opposite side 18, adjacent side 80, and hypotenuse 82. Find $\sin\theta$.

a. $\frac{41}{9}$

b. $\frac{3}{14}$

c. $\frac{8}{41}$

d. $\frac{9}{41}$ ✓

$\sin\theta = \frac{18}{82} = \frac{9}{41}$.

34 **[Intermediate]** • **Target: MCQ — Special angles**

[illegible] $\theta = \frac{\pi}{3}$ [illegible] In the unit circle, what are the Cartesian coordinates of the point corresponding to an angle $\theta = \frac{\pi}{3}$ radians?

a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ ✓

b. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

c. $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

d. $(0, 1)$

[illegible] (x, y) [illegible] θ [illegible] $x = \cos(\theta)$ ■ $y = \sin(\theta)$. [illegible] $\theta = \frac{\pi}{3}$ [illegible]: $\cos(\frac{\pi}{3}) = \frac{1}{2}$ $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$ [illegible] [illegible] $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. In the unit circle, the coordinates of the point (x, y) corresponding to an angle θ are given by $x = \cos(\theta)$ and $y = \sin(\theta)$. For the angle $\theta = \frac{\pi}{3}$ radians.

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35 [Intermediate] • Target: MCQ — Converting between radians and degrees

■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■ ABC■ ■■■■ ■■■■■■■■ ■■■■ B ■■ 90 ■■■■■■ ■■■■ ■■■■ ■■■■ ■■■■■■ AB ■■■■■■ 3 ■■ ■■■■■ ■■■■■■ BC ■■■■■■ 4 ■■■■ ■■■■ ■■ ■■■■ ■■■■■■ AC■ In a right-angled triangle ABC, where the angle at B is 90 degrees, if the length of side AB is 3 cm and the length of side BC is 4 cm, what is the length of the hypotenuse AC?

a. 5 ■■ ✓

b. 6 ■■

c. 7 ■■

d. 8 ■■

■■■■■■■ ■■■■■■ ■■■■■■■■■ ■■■■■■■ ■■■■ ■■■■■■ ■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■. ■■■■ ■■■■■■■■ ■■■■ ■■ ■■■■■ ■■■■ ■■■■■■ (AC) ■■■■■■ ■■■■■■ ■■■■■■ ■■■■■ ■■■■■■■■ ■■■■■■■■ (AB ■ BC). $AC^2 = AB^2 + BC^2$ ■■■■■ ■■■■■■■ ■■■■■■■■: $AC^2 = 3^2 + 4^2$ $AC^2 = 9 + 16$ $AC^2 = 25$ ■■■■■■■ ■■■■ AC■ ■■■■■ ■■■■■■ ■■■■■■■■■ ■■■■■ ■■■■■■■■: $AC = \sqrt{25}$ $AC = 5$ We use the Pythagorean theorem to determine the length of the hypotenuse in a right-angled triangle. The theorem states that the square of the length of the hypotenuse (AC) is equal to the sum of the sq.

36 [Intermediate] • Target: MCQ — The Pythagorean identity

■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■ ABC■ ■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■ B■ ■■■■ ■■■■ ■■■■ ■■■■■■ AB ■■■■■■ 3 ■■ ■■■■■ ■■■■■■ BC ■■■■■■ 4 ■■■■ ■■■■ ■■ ■■■■ ■■■■■■ AC■ In a right-angled triangle ABC, with the right angle at B, if the length of side AB is 3 cm and the length of side BC is 4 cm, what is the length of the hypotenuse AC?

a. 5 ■■ ✓

b. 7 ■■

c. 25 ■■

d. 12 ■■

■■■■■■■ ■■■■■■ ■■■■■■■■■ ■■■■■■ ■■■■ ■■■■■■ ■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■. ■■■■ ■■■■■■■■ ■■■■ ■■ ■■■■■ ■■■■ ■■■■■■ ■■■■■■ ■■■■■■ ■■■■■■ ■■■■■ ■■■■■■■■ ■■■■■■■■. ■■ ■■■■ ■■■■■■■■ $AC^2 = AB^2 + BC^2$. ■■■■■■■■■ ■■■■■■■ ■■■■■■■■: $AC^2 = 3^2 + 4^2$ $AC^2 = 9 + 16$ $AC^2 = 25$ ■■■■■ ■■■■■■ ■■■■■■■■■ ■■■■■■■■■ ■■■■■ ■■■■: $AC = \sqrt{25}$ $AC = 5$ ■■ ■■■■■■■ (■) ■■ ■■■■■■■■ ■■■■■■■■. We use the Pythagorean theorem to calculate the length of the hypotenuse in a right-angled triangle. The theorem states that the square of the hypotenuse's length is equal to.

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37 [Intermediate] • Target: MCQ — Simplifying trigonometric expressions

[illegible] $= 5$ [illegible] $= 3$. [illegible] In a right triangle, the hypotenuse is 5 and one leg is 3. Find the other leg.

a. 6

b. 1

c. 3

d. **4** ✓

[illegible] Trigonometry: $b = \sqrt{5^2 - 3^2} = \sqrt{16} = 4$ [illegible] 4. Applying Trigonometry principles: $b = \sqrt{5^2 - 3^2} = \sqrt{16} = 4$ Therefore, the correct choice is 4.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Graphs: amplitude and period • Cross-Lesson: 1.01 Algebra & Equations

[illegible] x In the right-angled triangle shown, what is the length of side x?

a. 10

b. **12** ✓

c. 14

d. 16

[illegible]. [illegible] 13. [illegible] 5 [illegible] x. [illegible]: $5^2 + x^2 = 13^2$ $25 + x^2 = 169$ [illegible] 25 [illegible]: $x^2 = 169 - 25$ $x^2 = 144$ [illegible]: $x = \sqrt{144}$ $x = 12$ We use the Pythagorean theorem in a right-angled triangle, which states that the square of the hypotenuse is equal to the sum of the squares of the other.

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39 [Challenging] • Target: MCQ — Graphs: phase shift and vertical shift • Cross-Lesson: 2.01 Functions & Transformations

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. **1** ✓

d. 5/3

The point P is on the unit circle, so its coordinates are $(\cos\theta, \sin\theta)$. We are given $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. The point Q has coordinates (x, y) and its distance from the origin is 5. The angle between the positive x-axis and OQ is α. Thus, $x = 5\cos\alpha$ and $y = 5\sin\alpha$. We are asked to find the maximum possible value of the product of the x-coordinates of P and Q, which is $\cos\theta \cdot x$. Substituting the given values, we h.

40 [Challenging] • Target: MCQ — Solving basic trigonometric equations • Cross-Lesson: 3.01 Sequences & Series

[redacted] 5 [redacted] 12. [redacted] In a right triangle, the two legs have lengths 5 and 12. Find the hypotenuse.

a. 12

b. 9

c. 15

d. **13** ✓

[redacted] Trigonometry: $c = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$ [redacted] 13. Applying Trigonometry principles: $c = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$ Therefore, the correct choice is 13.

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41 [Challenging] • Target: MCQ — Reference angles in other quadrants • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

■■ ■■■■■■ ABC■ ■■■■ ■■■■ $\angle A = 45^\circ$ ■ $\angle B = 60^\circ$ ■ ■■■■■ ■■■■■■ $a = 10$■ ■■■■ ■■ ■■■■ ■■■■■■ b ■■■■■■ In triangle ABC, if $\angle A = 45^\circ$, $\angle B = 60^\circ$, and the length of side $a = 10$, what is the exact length of side b?

a. $\frac{10\sqrt{3}}{\sqrt{2}}$

b. $\frac{10\sqrt{2}}{\sqrt{3}}$

c. $\frac{10\sqrt{6}}{2}$ ✓

d. $\frac{10\sqrt{3}}{2}$

■■■■■■■ ■■■■■■ ■■■■■■ ■■■■■ ■■■■ ■■■■ $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$. We use the Law of Sines, which states $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$. ■■■■■■ $A = 45^\circ$ ■ $B = 60^\circ$ ■ ■ $a = 10$. ■■■■■ ■■■■■■ b. We have $A = 45^\circ$, $B = 60^\circ$, and $a = 10$. We want to find b. ■■■■■■■ ■■■■■■■■: Applying the law: $\frac{10}{\sin 45^\circ} = \frac{b}{\sin 60^\circ}$ ■■■■■ ■■ $\sin 45^\circ = \frac{\sqrt{2}}{2}$ ■ $\sin 60^\circ = \frac{\sqrt{3}}{2}$. We know that.

42 [Challenging] • Target: MCQ — Complementary angle relationships • Cross-Lesson: 6.01 Limits & Derivatives

■■ ■■■■■■ ■■■■■■■■ ■■■■■■ ■■■■■ ABC ■■■■ $AB = 10$ ■ $AC = 15$. ■■■■■ ■■■■■■■ ■■■■■■■■■■ AD ■■■■■■■■ BAC■ ■■■■ ■■■■ D ■■■■ BC. ■■■■ ■■■■ $BD = 6$ ■■■■■ $BC = x$■ ■■■■■■ ■■■■■ x. In the attached figure, we have triangle ABC where $AB = 10$ and $AC = 15$. The line segment AD bisects angle BAC, with D lying on BC. If $BD = 6$ and the length of $BC = x$, find the value of x.

a. $x = 8$

b. $x = 9$ ✓

c. $x = 10$

d. $x = 12$

■■■■■■ ■■■■■■■ ■■■■■ ■■■■■■■■ ■■ ■■■■■■■■ ■■■■ ■■■■■ ■■■■■ ■■■■■■■■ ■■■■■■■■■■ ■■■■■■ ■■■■■■■■ ■■■■■■ ■■■■■ ■■■■■ ■■■■■■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■ ■■ ■■■■■■ ■■■■■■■■. ■■ ■■■■■■■ ABC■ ■■■■■■ AD ■■ ■■■■■ ■■■■■■■■ BAC. ■■■■■■ ■■■■■■■ $\frac{AB}{AC} = \frac{BD}{DC}$. ■■■■■■■■■ ■■■■■■■ ■■■■■■■■: $\frac{10}{15} = \frac{6}{DC}$ ■■■■■■■ ■■■■■■ ■■■■■■■■■: $10 \times DC = 15 \times 6$ $10 \times DC = 90$ $DC = \frac{90}{10}$ $DC = 9$ ■■■■ ■■■■■■ BC ■■ ■■■■■■ ■■■■■ BD ■ DC. $BC = BD + DC$ $x = 6 + 9$ $x = 15$ According to the Angle Bisector Theorem.

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43 [Challenging] • Target: MCQ — The unit circle: coordinates of key points • Cross-Lesson: 7.01 Integration & Applications

■■ ■■■■■■ ■■■■■■■■ ■■■■■■ ■■■■■ ABC ■■■ $AB = 10$ ■■■ $BC = 15$ ■■■ ■■■■■■ ■■■■■■■■ $B = 60^\circ$. ■■■ ■■ ■■■■ ■■■■■■■ ■■ ■■■■■■ A ■■■ ■■■■■■ BC ■■■■■■ BC ■■ ■■■■■■■ D■ ■■■■■■ ■■■■ ■■■■■■ AD. In the shown figure, we have a triangle ABC where $AB = 10$ cm, $BC = 15$ cm, and the measure of angle $B = 60^\circ$. If an altitude is drawn from vertex A to side BC, intersecting BC at point D, find the length of side AD.

a. $5\sqrt{3}$ ■■ ✓

b. $10\sqrt{3}$ ■■

c. $7.5\sqrt{3}$ ■■

d. $15\sqrt{3}$ ■■

■■■■■■■ ■■■■ ■■■■■■■■■ AD■ ■■■■■■■ ■■■■■■ ■■■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■ ABD. ■■■■■■ ■■■■■■ $AB = 10$ ■■ ■■■■■■■■■ $B = 60^\circ$. ■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■■ ■■■■■ ■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■ ■■■■■■ ■■■■■■ ■■■■ ■■■■■■■■■. ■■■■■ ■■■■■■: $\sin(B) = \frac{AD}{AB}$ $\sin(60^\circ) = \frac{AD}{10}$ ■■■■■ ■■ $\sin(60^\circ) = \frac{\sqrt{3}}{2}$. ■■■■■■■■■ ■■ ■■■■■■■■■: $\frac{\sqrt{3}}{2} = \frac{AD}{10}$ ■■■■ AD■ ■■■■■ ■■■■■■■■ ■■ 10: $AD = 10 \times \frac{\sqrt{3}}{2}$ $AD = 5\sqrt{3}$ ■■. To find the length of the altitude AD, we can cons.

44 [Challenging] • Target: MCQ — Word problems: angles of elevation • Cross-Lesson: 8.01 Exponentials & Logarithms

■■ ■■■■■■ ■■■■■■■■■ ■■■■■■■ ABC ■■ ■■■■■■ ■■■■■■ $AB = 10$■ $AC = 15$■ ■■■■■■■■ ■■■■■■■■■■ AD ■■■■■ ■■■■■■■■ A■ ■■■■ D ■■■■ ■■■■ BC. ■■■■ ■■■■ ■■■■ $BD = 6$■ ■■■■■■ ■■■■ DC. In the given figure, triangle ABC has side lengths $AB = 10$, $AC = 15$, and segment AD bisects angle A, with D on BC. If the length of $BD = 6$, find the length of DC.

a. **8** ✓

b. 9

c. 10

d. 12

■■■■■■ ■■■■■■■ ■■■■■ ■■■■■■■■■■ ■■■■ ■■■■■ ■■■■■ ■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ ■■■■■ ■■ ■■■■■■■■ ■■■■■■ ■■■■■ ■■■■■ ■■■■■■ ■■■■■■■■■. ■■ ■■■■■■■ ABC■ AD ■■■■■ ■■■■■■■■ A■ ■■■■ ■■■■■■: $\frac{AB}{AC} = \frac{BD}{DC}$. ■■■■■■■ ■■■■■■ ■■■■■■■■■: $\frac{10}{15} = \frac{6}{DC}$. ■■■■■ ■■■■■■: $\frac{2}{3} = \frac{6}{DC}$. ■■■■ DC■ ■■■■■ ■■■■■■■■ ■■ ■■■■■■■■: $2 \cdot DC = 3 \cdot 6$. $2 \cdot DC = 18$. ■■■■■ ■■■■■■■■ ■■■■ 2: $DC = \frac{18}{2} = 9$. According to the Angle Bisector Theorem, the ratio of the lengths of the two sides that form the bise.

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45 [Challenging] • Target: Right-triangle trigonometric ratios • Cross-Lesson: 9.01 Statistics & Probability

■■ ■■■■ ABC ■ ■■■ ■■■ $a = 5$■ $b = 7$■ ■■■■■■■■ $C = 60^\circ$ ■ ■■■ ■■■■ ■■■ ■■■■■ c■

In triangle ABC, if $a = 5$, $b = 7$, and angle $C = 60^\circ$, what is the length of side c?

a. $\sqrt{39}$

b. $\sqrt{49}$

c. $\sqrt{61}$ ✓

d. $\sqrt{75}$

■■■■■■ ■■■■■ ■■■ ■■■■■■ ■■■■■ ■■■ ■■■■■ c ■■ ■■■■■■. ■■■■■ ■■■ ■■■■■■ ■■■ ■■■ ■■ $c^2 = a^2 + b^2 - 2ab\cos(C)$. ■■■■ ■■■■■■ ■■■■■■■: $a = 5$■ $b = 7$■ ■ $C = 60^\circ$. ■■■■ ■■ $\cos(60^\circ) = \frac{1}{2}$. ■■■■■ $c^2 = 5^2 + 7^2 - 2(5)(7)(\frac{1}{2})$. $c^2 = 25 + 49 - 35$. $c^2 = 74 - 35$. $c^2 = 39$. ■■■■■ $c = \sqrt{39}$. We use the Law of Cosines to calculate the length of side c \) in the triangle. The Law of Cosines states that $c^2 = a^2 + b^2 - 2ab\cos(C)$. Substitute the given values.

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